

國立清華大學課程大綱

科號		組別		學分	3	人數限制	80
上課時間	MnM5M6			教室	TBA		
科目中文名稱	數據分析與機器學習						
科目英文名稱	Data Analysis and Machine Learning						
任課教師	余朝恩						

一、課程說明	This course surveys basic concepts and technical tools required for proceeding the Bayesian data analysis. The goal of this course targets to equip students with the understandings of how to interpret and what are the limits of common data analytical results. In addition to introducing theoretically analytical models, proper programming skills are covered in this course. 因為在本課程獲得高分的學長姊陸續在 ASML 和台灣微軟公司、玉山銀行 Fintech 部門獲得實習機會並轉正職。故本課程為普及機器學習教學，不需要學生有任何修課背景。需要用到的統計、微積分、矩陣代數等，乃至於 Python 程式設計，都會給同學們教學/複習一次。 Note: This course requires learning Python programming skills for verifying theoretical methods and practicing analytical applications.
二、指定用書	Doing Bayesian Data Analysis, 2nd Ed. (ISBN: 0124058884) A First Course in Machine Learning (ISBN: 9781498738484)
三、教學方式	Lecture: 30% Discussion: 30% Practices: 40%
五、教學進度	Week 1: The history of Bayesian statistics

	Week 2: Introduction of Python programming Week 3: Review of foundations in probability Week 4: Bayes theorem and applications Week 5-6: Bayesian vs frequentist inference Week 7: Bayesian inference with Python Week 8-10: MCMC simulations Week 11: MCMC simulations with Python Week 12-13: Bayes classifier Week 14-15: Bayesian regression and model selection Week 16: A comparison between Bayesian and econometric analyses
六、成績考核	Exams 40%; Quizzes 60%
七、講義位址 http://	http://mx.nthu.edu.tw/~chaoenyu/